4. Experimental Setup & Benchmark Protocol

Target Venues: *Automation in Construction* (Elsevier, IF: 10.3) | *Advanced Engineering Informatics* (Elsevier, IF: 8.0)
Companion Documents: `paper_idea.md`, `03_methodology.md`, `benchmark_cases.json`

4.1. Benchmark Suite Design Principles & Scope

To rigorously test the proposed neuro-symbolic framework and falsify Hypotheses H1–H3, an experimental benchmark suite of **24 structural design instances** was constructed. The suite reflects real-world structural engineering workflows spanning gravity load takedown, modal eigenvalue dynamics, Equivalent Lateral Force (ELF) seismic procedures, multi-directional Response Spectrum Analysis (RSA), direct earthquake accelerogram time-history numerical integration, multimodal drawing ingestion, and production AutoCAD `.dxf` drafting with Bar Bending Schedules (BBS).

In alignment with the core capabilities of the underlying symbolic solver ecosystem, all environmental lateral loading is strictly focused on **earthquake engineering** per ASCE 7-16/22 and codified concrete capacity checks per ACI 318-19/25. Unverified environmental loads (such as wind or snow) are excluded from the test matrix to maintain absolute ground-truth determinism against commercial CAE reference software (CSI SAP2000 and ETABS).

The 24 benchmark cases are divided into three progressive groups:

1. **Group A: 2D Plane Frames (Cases A01–A08)**: Evaluates fundamental 2D frame mechanics, tributary gravity load distribution, modal vibration periods, 2D Response Spectrum Analysis under both ASCE 7 code spectra and recorded ground motion spectra (1994 Northridge), and single-bay self-healing redesign reflection.
2. **Group B: 3D Space Frames (Cases B01–B08)**: Evaluates 3D direct stiffness finite element assembly, multi-story biaxial gravity distribution, ASCE 7 Equivalent Lateral Force (ELF) seismic procedures with orthogonal combination ($1.0E_X \pm 0.3E_Y$), 3D coupled translational-torsional modal dynamics, 3D Response Spectrum Analysis, plan-asymmetric torsional irregularity, and 3D seismic self-healing redesign.
3. **Group C: Advanced Dynamic, Multimodal Vision, CAE & CAD Pipelines (Cases C01–C08)**: Evaluates direct numerical time-history integration of nonlinear/linear equations of motion under recorded accelerograms (1940 El Centro, 1994 Northridge), lossless round-trip CAE exchange with CSI SAP2000 (`.s2k`) and ETABS (`.e2k`), multimodal architectural floor plan ingestion via Vision-Language Models (`struct_agent.vision`), dynamic reflection under time-history demands, and autonomous end-to-end seismic design-to-CAD production.

4.2. Systematic Benchmark Case Taxonomy (24 Cases)

Table 2 presents the systematic taxonomy of all 24 benchmark problems, detailing the structural archetype, loading regime, governing code checks, expected symbolic tools, and verified baseline metrics.

Table 2: Standardized 24-Case Structural and Seismic Benchmark Matrix

Case ID	Name & Structural Archetype	Geometric Configuration	Loading & Analysis Regime	Design Required?	Verified Baseline ($V_{\text{base}}, \text{DCR}_{\text{base}}$)	Expected Tool Pipeline
A01	Single-Bay Portal Frame (Office)	1 bay, 1 story; $L = 6.0\ \text{m}, H = 3.5\ \text{m}$	Office Gravity ($w_{\text{trib}} = 4.0\ \text{m}$); Linear Static	Yes	$1.757\ \text{m}^3$ (DCR 0.846)	`create_frame_model`, `apply_gravity_loads`, `run_analysis`, `run_design`, `generate_drawings`
A02	Wide-Span Portal Frame (Storage)	1 bay, 1 story; $L = 8.0\ \text{m}, H = 4.0\ \text{m}$	Heavy Storage Gravity ($w_{\text{trib}} = 4.5\ \text{m}$); Linear Static	Yes	$3.820\ \text{m}^3$ (DCR 0.864)	`create_frame_model`, `apply_gravity_loads`, `run_analysis`, `run_design`
A03	2-Bay 2-Story Frame (Office)	2 bays, 2 stories; $L = 5.0\ \text{m}, H = 3.2\ \text{m}$	Office Gravity ($w_{\text{trib}} = 5.0\ \text{m}$); Linear Static	Yes	$5.772\ \text{m}^3$ (DCR 0.599)	`create_frame_model`, `apply_gravity_loads`, `run_analysis`, `run_design`
A04	2-Bay 3-Story Frame (Residential)	2 bays, 3 stories; $L = 6.0\ \text{m}, H = 3.0\ \text{m}$	Residential Gravity ($w_{\text{trib}} = 4.5\ \text{m}$); Linear Static	Yes	$9.720\ \text{m}^3$ (DCR 0.677)	`create_frame_model`, `apply_gravity_loads`, `run_analysis`, `run_design`
A05	2D Modal Dynamics Frame	2 bays, 3 stories; $L = 6.0\ \text{m}, H = 3.5\ \text{m}$	Consistent Mass Formulation; Generalized Eigenproblem	No	Pure Dynamics (T_1, T_2, T_3)	`create_frame_model`, `run_modal_analysis`, `inspect_modal_results`
A06	2D Response Spectrum (ASCE 7)	2 bays, 3 stories; $L = 6.0\ \text{m}, H = 3.5\ \text{m}$	ASCE 7 Spectrum ($S_{DS} = 1.0g, S_{D1} = 0.6g$); CQC Modal Combination	Yes	$13.309\ \text{m}^3$ (DCR 0.880)	`create_frame_model`, `apply_gravity_loads`, `run_response_spectrum`, `run_design`
A07	2D RSA Ground Motion (Northridge)	3 bays, 4 stories; $L = 5.5\ \text{m}, H = 3.5\ \text{m}$	1994 Northridge Accelerogram Response Spectrum; CQC	Yes	$26.705\ \text{m}^3$ (DCR 0.880)	`create_frame_model`, `apply_gravity_loads`, `run_response_spectrum`, `run_design`

Case ID	Name & Structural Archetype	Geometric Configuration	Loading & Analysis Regime	Design Required?	Verified Baseline ($V_{\text{base}}, \text{DCR}_{\text{base}}$)	Expected Tool Pipeline
A08	2D Overstressed Reflection Portal	1 bay, 1 story; $L = 7.0\ \text{m}, H = 3.5\ \text{m}$	Heavy Storage ($w_{\text{trib}} = 6.0\ \text{m}$); Initial $\text{DCR}_0 > 1.0$	Yes	$3.710\ \text{m}^3$ (DCR 0.797)	`create_frame_model`, `run_analysis`, `run_design`, `modify_project`, `generate_drawings`
B01	3D Hostel Block Space Frame	2×2 bays, 2 stories; $L_x = 6.0\ \text{m}, L_y = 6.0\ \text{m}, H = 3.5\ \text{m}$	Residential Floor Gravity; 3D Direct Stiffness	No	Kinematics Verification	`create_frame_model`, `apply_gravity_loads`, `run_analysis`
B02	3D 3×2 Bay Commercial Frame	3×2 bays, 3 stories; $L_x = 5.0\ \text{m}, L_y = 6.0\ \text{m}, H = 3.2\ \text{m}$	Office Gravity ($w_{\text{trib}} = 5.0\ \text{m}$); Biaxial Moments	Yes	$85.104\ \text{m}^3$ (DCR 0.846)	`create_frame_model`, `apply_gravity_loads`, `run_analysis`, `run_design`
B03	3D Seismic ELF Space Frame	2×2 bays, 3 stories; $L_x = 6.0\ \text{m}, L_y = 6.0\ \text{m}, H = 3.5\ \text{m}$	ASCE 7 Seismic ELF ($S_{DS} = 1.2g, S_{D1} = 0.5g, R = 5$); 26 LRFD Combinations	Yes	$143.100\ \text{m}^3$ (DCR 0.983)	`create_frame_model`, `apply_gravity_loads`, `apply_lateral_loads`, `run_analysis`, `run_design`
B04	3D Modal Eigenvalue Dynamics	2×2 bays, 3 stories; $L_x = 6.0\ \text{m}, L_y = 6.0\ \text{m}, H = 3.5\ \text{m}$	3D Mass Matrix; Coupled Translational-Torsional Modes	No	3D Dynamics (T_1, T_2, T_3)	`create_frame_model`, `run_modal_analysis`, `inspect_modal_results`
B05	3D Response Spectrum (X-Direction)	2×2 bays, 3 stories; $L_x = 6.0\ \text{m}, L_y = 6.0\ \text{m}, H = 3.5\ \text{m}$	ASCE 7 Spectrum ($S_{DS} = 1.1g, S_{D1} = 0.55g$); 3D CQC Combination	Yes	$116.640\ \text{m}^3$ (DCR 0.929)	`create_frame_model`, `apply_gravity_loads`, `run_response_spectrum`, `run_design`
B06	3D Multi-Directional Dynamic RSA	3×2 bays, 3 stories; $L_x = 5.0\ \text{m}, L_y = 6.0\ \text{m}, H = 3.2\ \text{m}$	1940 El Centro Multi-Directional Spectrum ($1.0X \pm 0.3Y$)	Yes	$107.700\ \text{m}^3$ (DCR 0.948)	`create_frame_model`, `apply_gravity_loads`, `run_response_spectrum`, `run_design`
B07	3D High-Seismic Reflection Frame	2×2 bays, 2 stories; $L_x = 6.0\ \text{m}, L_y = 6.0\ \text{m}, H = 3.5\ \text{m}$	High Seismic ELF ($S_{DS} = 1.2g$); Initial $\text{DCR}_0 = 1.34 > 1.0$	Yes	$56.857\ \text{m}^3$ (DCR 0.626)	`create_frame_model`, `apply_lateral_loads`, `run_analysis`, `run_design`, `modify_project`
B08	3D Asymmetric Torsional Frame	3 stories; Non-symmetric perimeter stiffness	ASCE 7 ELF with Accidental Torsional Eccentricity (5%)	Yes	$113.636\ \text{m}^3$ (DCR 0.940)	`create_frame_model`, `apply_gravity_loads`, `apply_lateral_loads`, `run_analysis`, `run_design`
C01	Dynamic Time-History (El Centro)	2 bays, 3 stories; $L = 6.0\ \text{m}, H = 3.5\ \text{m}$	1940 El Centro Accelerogram ($a_{\text{peak}} = 0.348g$); Newmark-Beta Integration	Yes	$20.869\ \text{m}^3$ (DCR 0.951)	`create_frame_model`, `apply_gravity_loads`, `run_time_history`, `inspect_dynamic_results`, `run_design`
C02	3D Dynamic Time-History (Northridge)	2×2 bays, 2 stories; $L_x = 6.0\ \text{m}, L_y = 6.0\ \text{m}, H = 3.5\ \text{m}$	1994 Northridge Accelerogram ($a_{\text{peak}} = 0.583g$); Step-by-Step Dynamics	No	Linear Dynamics Verification	`create_frame_model`, `run_time_history`, `inspect_dynamic_results`
C03	CSI SAP2000 Round-Trip CAE	Single-bay portal frame (`sample_frame.s2k`)	Linear Static FEA & ACI 318 Checks; CAE Exchange	Yes	$1.710\ \text{m}^3$ (DCR 0.846)	`import_from_sap2000`, `validate_model`, `run_analysi` `run_design`, `export_to_sap20`
C04	CSI ETABS Round-Trip CAE	Multi-story space frame (`sample_frame.e2k`)	Multi-story Connectivity & Boundary Restraints Validation	No	Round-Trip Geometry Parity	`import_from_etabs`, `validate_model`, `run_analysi` `export_to_etabs`
C05	Multimodal Vision Floor Plan Ingestion	3-story educational facility (`floor_plan_3story_school.PNG`)	Architectural Plan to 3D Frame Compilation (`create_model_from_plan_draft`)	No	Topological Topology Parity	`create_model_from_plan_dra` `validate_model`, `inspect_project`
C06	Multimodal Vision-to-Seismic Design	4-story commercial facility (`mall_plan_4story.PNG`)	Multimodal Compilation $\rightarrow$ Seismic ELF ($S_{DS} = 1.2g, R = 5$) $\rightarrow$ ACI 318	Yes	$92.664\ \text{m}^3$ (DCR 0.866)	`create_model_from_plan_dra` `apply_gravity_loads`, `apply_lateral_loads`, `run_analysis`, `run_design`
C07	Dynamic Reflection (El	2 bays, 3 stories; Initial $0.30 \times 0.30\ \text{m}$ columns	1940 El Centro Dynamic Record; Initial $\text{DCR}_0 = 1.38 > 1.0$	Yes	$20.869\ \text{m}^3$ (DCR 0.951)	`create_frame_model`, `run_time_history`, `run_desig`

Case ID	Name & Structural Archetype	Geometric Configuration	Loading & Analysis Regime	Design Required?	Verified Baseline ($V_{\text{base}}, \text{DCR}_{\text{base}}$)	Expected Tool Pipeline
	Centro)					`modify_project`
C08	Autonomous Seismic Design-to-CAD	2 bays, 2 stories; $L = 5.0\ \text{m}, H = 3.2\ \text{m}$	Response Spectrum ($S_{DS} = 1.0g, S_{D1} = 0.6g$) → ACI 318 → AutoCAD `.dxf`	Yes	$6.072\ \text{m}^3$ (DCR 0.846)	`create_frame_model`, `run_response_spectrum`, `run_design`, `generate_drawin` `inspect_detailing_results`

4.3. Real-World Architectural Benchmark Fixtures

To validate multimodal perceptual compilation under realistic architectural layouts, three real-world building floor plans are incorporated as benchmark fixtures:

1. **3-Story Reinforced Concrete School Facility (`floor_plan_3story_school.PNG`)**:
   - *Case Binding*: Case C05 (`C05_Vision_FloorPlan_Draft_Ingestion`).
   - *Structural Layout*: 3 stories at $3.5\ \text{m}$ story height, framing an educational facility with overall footprint $21.5\ \text{m} \times 9.75\ \text{m}$.
   - *Grid Topology*: Orthogonal structural grids Grids A, B, C, D along the longitudinal axis and Grids 1, 2, 3, 4 along the transverse axis. The framing defines standard classroom modules with a central longitudinal circulation corridor.
   - *Role in Benchmarking*: Assesses the Vision-Language Model's ability to interpret structural grid notations, bay spacings, wall/column symbols, and correctly compile a valid, watertight 3D frame model (`FloorPlanDraft` → `struct_core.Project`).
2. **2-Story Residential Hostel Block (`hostel_block_plan_2 story.PNG`)**:
   - *Case Binding*: Case B01 (`B01_3D_2x2_2Story_Office`).
   - *Structural Layout*: 2 stories at $3.5\ \text{m}$ story height, framing a residential block with overall footprint $27.3\ \text{m} \times 10.9\ \text{m}$.
   - *Grid Topology*: Multi-bay modular grid system (Grids A through H along the longitudinal axis and Grids 1 through 3 along the transverse axis) with repetitive room units, perimeter columns, and dual-direction framing.
   - *Role in Benchmarking*: Provides the geometric blueprint for multi-bay 3D space frame direct stiffness analysis, validating 3D gravity load takedown and tributary surface load decomposition.
3. **4-Story Commercial Mall Complex (`mall_plan_4story.PNG`)**:
   - *Case Binding*: Case C06 (`C06_Vision_To_Seismic_Design_Pipeline`).
   - *Structural Layout*: 4 stories at $3.2\ \text{m}$ story height, framing a commercial retail complex with overall footprint $49'\text{-}0'' \times 36'\text{-}0''$ ($14.94\ \text{m} \times 10.97\ \text{m}$).
   - *Grid Topology*: Grids 5, 4, 3, 2, 1 along the longitudinal direction and Grids A, B, C, D along the transverse direction, accommodating large open retail spans and centralized column clusters.
   - *Role in Benchmarking*: Forms the ultimate end-to-end stress test: extracting architectural geometry from raster pixels → compiling a 3D structural frame → applying commercial gravity and high-seismic ASCE 7 ELF lateral forces ($S_{DS} = 1.2g, S_{D1} = 0.5g, R = 5.0$) → executing 3D finite element analysis → verifying ACI 318 biaxial member capacities.

4.4. Comparative Baselines & Ablation Configurations

To evaluate the contribution of each architectural component, four comparative configurations are executed across the benchmark suite:

1. **Baseline 1 (Direct Zero-Shot LLM Prompting — `direct_llm`)**:
   - The language model receives natural language structural problem specifications and must directly estimate internal forces, member sizing, and capacity ratios without access to external symbolic tools.
   - *Role*: Quantifies the baseline Calculation Hallucination Rate (CHR) and proves that pure neural reasoning violates physical equilibrium in indeterminate structures.
2. **Baseline 2 (Code-Generation LLM — `code_gen`)**:
   - The language model is prompted to generate standalone executable Python scripts using general numerical libraries (NumPy, SciPy) to model and solve the structural problem.
   - *Role*: Evaluates whether raw code generation without a domain-specific schema or state-machine guardrails can reliably solve complex 3D seismic mechanics without syntax, singularity, or code-compliance failures.
3. **Baseline 3 (Ablated Agent without Reflection — `ablated`)**:
   - The agent is equipped with the complete suite of symbolic tools (`fem2d`, `fem3d`, `structdyn`, `aci318`, `struct_draw`), but the closed-loop self-healing reflection mechanism is disabled. If an initial trial design yields $\text{DCR}_{\text{gov}} > 1.0$, execution terminates immediately as non-compliant.
   - *Role*: Directly isolates the empirical value of the closed-loop diagnostic reflection state machine on Code Compliance Rate (CCR).
4. **Proposed Framework (Full Neuro-Symbolic Agent with Reflection — `full_agent`)**:
   - The complete architecture featuring ontology-grounded validation (`struct_core`), symbolic mechanics solvers, multimodal drawing compilation (`struct_agent.vision`), and closed-loop self-healing redesign driven by DCR_{gov} diagnostic feedback.

4.5. Multi-Model Evaluation Testbed

To test cross-model generalization (Hypothesis H3), benchmarks are evaluated across four prominent Large Language Model backbones representing both proprietary frontier APIs and local/cloud open-weight architectures:

Model Identifier	Architecture & Provider	Parameter Scale	Deployment Modality	Context Window
`gemini-3.8-flash`	Frontier Closed-Source (Google DeepMind)	Proprietary	Cloud API (`google-genai` SDK)	1,000,000 tokens
`qwen3.5:9b`	Open-Weight Dense (Alibaba Cloud)	9B	Local Workstation via Ollama	32,768 tokens
`gpt-oss:120b-cloud`	Open-Weight MoE/Dense Foundation Model	120B	High-Performance Cloud Host via Ollama	65,536 tokens
`nemotron-3-ultra:cloud`	Open-Weight Math/Engineering-Tuned (NVIDIA)	Ultra Scale	High-Performance Cloud Host via Ollama	32,768 tokens

4.6. Quantitative Evaluation Metrics

The experimental evaluation employs five rigorous, mathematically formulated metrics:

1. Task Completion Rate (TCR)

$$\mathrm{TCR} = \frac{N_{\text{completed}}}{N_{\text{total}}} \times 100\% \quad (N_{\text{total}} = 24)$$

A benchmark run is deemed completed ($N_{\text{completed}}$) if and only if the agent successfully synthesizes or imports a valid `struct_core.Project` passing the symbolic guard $\mathcal{V}(\mathcal{S}) = 1$ and executes all required pipeline tools in `expected_tools` without unhandled runtime exceptions.

2. Code Compliance Rate (CCR)

$$\mathrm{CCR} = \frac{N_{\text{compliant}}}{N_{\text{design}}} \times 100\% \quad (N_{\text{design}} = 18)$$

Evaluated strictly across the 18 design-mandated benchmark cases. A completed design is code-compliant if and only if the governing Demand-Capacity Ratio satisfies:

$$\mathrm{DCR}_{\text{gov}} = \max_{e \in \mathcal{E},\, c \in \mathcal{C}} \left\{ \frac{P_{u,e,c}}{\phi P_{n,e}},\ \frac{M_{u,e,c}}{\phi M_{n,e}},\ \frac{V_{u,e,c}}{\phi V_{n,e}} \right\} \leq 1.00$$

3. Calculation Hallucination Rate (CHR)

$$\mathrm{CHR} = \frac{N_{\text{hallucinated_claims}}}{N_{\text{total_numerical_claims}}} \times 100\%$$

A reported numerical engineering claim (e.g., base shear V_b, fundamental period T_1, peak bending moment M_u, or governing DCR) is classified as a calculation hallucination if it deviates by $> 5.0\%$ from the exact deterministic mechanics solution. In the proposed framework, $\mathrm{CHR} = 0.0\%$ by architectural construction.

4. Self-Healing Convergence Iterations (N_{iter})

For benchmark cases initialized with undersized trial cross-sections ($\mathrm{DCR}_{\text{gov}}^{(0)} > 1.0$), N_{iter} quantifies the number of closed-loop reflection cycles (`modify_project` $\rightarrow$ `run_analysis` $\rightarrow$ `run_design`) required to achieve structural feasibility ($\mathrm{DCR}_{\text{gov}}^{(N_{\text{iter}})} \leq 1.0$).

5. Volumetric Material Economy Ratio (η_{vol})

$$\eta_{\text{vol}} = \frac{V_{\text{concrete}}^{\text{agent}}}{V_{\text{concrete}}^{\text{baseline}}}$$

where $V_{\text{concrete}} = \sum_{e \in \mathcal{E}} A_e L_e$ is the total structural concrete volume (m^3) of the converged design, and $V_{\text{concrete}}^{\text{baseline}}$ is the verified human engineering baseline volume established by `generate_human_baselines.py`. For cases incorporating full AutoCAD reinforcement drafting (`struct_draw`), the composite material cost metric is additionally inspected:

$$\eta_{\text{material}} = \frac{V_{\text{concrete}}^{\text{agent}} + \alpha W_{\text{steel}}^{\text{agent}}}{V_{\text{concrete}}^{\text{baseline}} + \alpha W_{\text{steel}}^{\text{baseline}}}$$

where W_{steel} is the total rebar mass (tonnes) extracted from the Bar Bending Schedule, and $\alpha \approx 60$–80 is the regional unit cost ratio of structural steel to concrete.